

DEV_IMPL_GUIDE - DD: WO-01-FRAMEWORK — Work Order Module Framework

1. Purpose

This guide explains how to implement DD_WO-01-FRAMEWORK-v1.0.md as executable SOPHIX behavior.

Use it with:

- DD_WO-01-FRAMEWORK-v1.0.md
- DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md when this DD participates in subscription workflow operations.
- DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md when this DD references worker topics.

2. Runtime Actors

Runtime component	Responsibility
API pod / module service	Receives requests, validates, loads authoritative state, starts commands.
Owning module database	Stores tables/configuration owned by this DD.
Reference modules	Provide data through approved APIs/client wrappers.
Camunda	Runs BPMN process when this DD is workflow-driven.
Worker apps	Execute logical worker topics.
Drools	Evaluates packages when rules are declared.
Kafka/outbox	Publishes success/failure and integration events.

3. Before Implementation

Confirm these contracts from the DD:

Contract	Value
Endpoint scan reference	GET /api/customers/{id}
BPMN/process key	configured BPMN process
Drools package	configured Drools package
Tables	wo_equipment_ref_stub
Events	WorkOrderAssigned, WorkOrderAttachmentAdded, WorkOrderCancellationRequested, WorkOrderCancelled, WorkOrderClaimedByDispatcher, WorkOrderClaimedByTechnician, WorkOrderCreated, WorkOrderDispatched, WorkOrderEscalationCandidate, WorkOrderFinalizationPending, WorkOrderFinalized, WorkOrderInProgress, WorkOrderNoteAppended, WorkOrderPhaseTransitioned, WorkOrderReassigned, WorkOrderUserActionReceived

4. Execution Flow

1. Apply migrations and seed configuration/catalog rows.
2. Implement the module service/repository layer around the tables and invariants.
3. Implement public/internal APIs from the DD.
4. Implement rule evaluation if the DD defines Drools packages.
5. Implement event production/consumption through outbox where the DD defines events.
6. Add integration tests for table constraints, API validation, and event payloads.

5. API And Command Handling

Endpoints found in the DD:

- GET /api/customers/{id}
- GET /api/work-orders
- GET /api/work-orders/wo_01HZ...
- GET /api/work-orders/wo_01HZ.../print-pdf
- GET /api/work-orders/{woId}
- GET /api/work-orders/{woId}/print-pdf
- GET /api/work-orders?operatorCode=WIK&status=PENDING, ASSIGNED&kind=SUPPORT&techRegionCode=NRB-KAREN&limit=50
- GET /engine-rest/process-instance/{id}/variables/{var}
- POST /api/work-orders
- POST /api/work-orders/wo_01HZ.../assign
- POST /api/work-orders/wo_01HZ.../attachments
- POST /api/work-orders/wo_01HZ.../cancel
- POST /api/work-orders/wo_01HZ.../claim
- POST /api/work-orders/wo_01HZ.../finalize-first-confirm
- POST /api/work-orders/wo_01HZ.../finalize-revert
- POST /api/work-orders/wo_01HZ.../notes
- POST /api/work-orders/wo_01HZ.../start
- POST /api/work-orders/wo_01HZ.../transition-phase
- POST /api/work-orders/wo_01HZ...T2/assign
- POST /api/work-orders/wo_01HZ...T2/claim
- POST /api/work-orders/wo_01HZ...T3/assign
- POST /api/work-orders/wo_01HZ...T3/claim
- POST /api/work-orders/wo_01HZ...T3/start
- POST /api/work-orders/wo_01HZ...Z2/assign
- POST /api/work-orders/wo_01HZ...Z2/start
- POST /api/work-orders/{id}/assign
- POST /api/work-orders/{id}/cancel
- POST /api/work-orders/{id}/claim
- POST /api/work-orders/{id}/equipment-bindings
- POST /api/work-orders/{id}/finalize-first-confirm
- POST /api/work-orders/{id}/finalize-revert
- POST /api/work-orders/{id}/finalize-second-confirm
- POST /api/work-orders/{id}/reassign
- POST /api/work-orders/{id}/start
- POST /api/work-orders/{id}/transition-phase
- POST /api/work-orders/{woId}/assign
- POST /api/work-orders/{woId}/attachments
- POST /api/work-orders/{woId}/cancel
- POST /api/work-orders/{woId}/claim
- POST /api/work-orders/{woId}/finalize-first-confirm

- POST /api/work-orders/{woId}/finalize-revert
- POST /api/work-orders/{woId}/finalize-second-confirm
- POST /api/work-orders/{woId}/notes
- POST /api/work-orders/{woId}/reassign
- POST /api/work-orders/{woId}/start
- POST /api/work-orders/{woId}/transition-phase
- POST /engine-rest/deployment/create
- POST /engine-rest/external-task/fetchAndLock
- POST /engine-rest/message
- POST /engine-rest/process-definition/key/WorkOrder_Installation_WIK/start
- POST /start

Implementation rules:

- Resolve actor user and role from auth context, not from public request body.
- Check idempotency before inserting a new operation or creating a side effect.
- Return the existing operation/result for the same idempotency key and same request hash.
- Return 409 IDEMPOTENCY_KEY_REUSED for the same key with a different request hash.
- Use transactions or unique constraints for concurrency-sensitive creation.

6. Data Loading

Load data in this order:

1. Authoritative aggregate state from the owning module.
2. Operator/module config from this DD's config tables.
3. Catalog/reference data from approved APIs or client wrappers.
4. Current in-flight operation state from the framework, where relevant.

Do not read current mutable state from cache for state-changing operations.

7. Drools Configuration

Packages found:

- No Drools package explicitly declared.

For each package:

- Define fact classes that mirror the request, current state, config, and reference data.
- Keep rule output as explicit decision fields.
- Unit-test accept and reject cases from the DD.
- BPMN should re-check critical rules after start if state may have changed.

8. BPMN And Worker Topics

Worker topics found:

- await-contractor-dispatcher-pick
- await-finalize-second-confirm
- await-tech-start
- fulfillment-order-capture-worker-7
- order-create-install-wo

- wait-install-wo-finalized
- wo-auto-assign-contractor
- wo-emit-state-event
- wo-enforce-finalize-checklist
- wo-mark-cancelled
- wo-shifting-update-address-on-finalize
- wo-validate-create

Implementation rules:

- Model service tasks with `camunda:type="external"` and the exact topic string.
- Worker handler must validate required variables before calling downstream services.
- Store outputs as process variables with stable names.
- Use message correlation keys exactly as specified by the DD.
- Add compensation paths for each external side effect.

9. Events

Events found:

- WorkOrderAssigned
- WorkOrderAttachmentAdded
- WorkOrderCancellationRequested
- WorkOrderCancelled
- WorkOrderClaimedByDispatcher
- WorkOrderClaimedByTechnician
- WorkOrderCreated
- WorkOrderDispatched
- WorkOrderEscalationCandidate
- WorkOrderFinalizationPending
- WorkOrderFinalized
- WorkOrderInProgress
- WorkOrderNoteAppended
- WorkOrderPhaseTransitioned
- WorkOrderReassigned
- WorkOrderUserActionReceived

Event rules:

- Emit success only after authoritative commit.
- Emit rejected/failure event only after compensation/final failure decision.
- Use the outbox table/dispatcher; do not publish directly inside the transaction boundary unless the platform standard allows it.

10. Smoke Tests

Add tests for:

- Happy path.
- Invalid role or trigger.

- Invalid state.
- Missing config/catalog row.
- Dependency failure.
- Retry/idempotency.
- Event payload correctness.
- Worker topic failure and compensation.

11. Common Mistakes

- Treating worker-topic names as shell commands.
- Querying another module's database directly.
- Reading current state from cache in a state-changing flow.
- Emitting success before final commit.
- Hiding manual recovery paths for partial external side effects.
- Creating one deployment per topic when a consolidated worker app would be easier to operate.